

Shubhan Chari

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EDUCATION

University of Illinois at Urbana-Champaign – GPA: 4.00 / 4.00 Expected May 2028
B.S. in Statistics & Computer Science with Honors / Minors in Data Science & Informatics Urbana, IL
• **Relevant Coursework:** Algorithms & Models of Computation, Data Structures (C++), Intro to Computer Systems (C), Modeling & Learning in Data Science (Python), Statistical Modeling (R), Intro to Databases (SQL), Reading & Writing Data

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, C++, Java, R, Go, TypeScript (React), PyTorch, scikit-learn, Pandas, Numpy, Excel
Technologies: MongoDB, PostgreSQL, Docker, AWS, Github Actions, Jenkins, Git, HuggingFace, Streamlit, Vercel, Render
Concepts: DSA, Computer Vision, NLP, Neural Networks, REST API, XGBoost, ARIMA, ETL, FastAPI, OpenAI CLIP

EXPERIENCE

Kencor Health Aug 2026 – Present
Machine Learning Engineer Intern Urbana, IL

- Built a full-stack blood pressure dashboard and graphic animation (Java REST API with RLS, MongoDB, AWS, React/Plotly) delivering 12-month trend analysis, hypertension staging, and outlier detection to care teams at 68 clinics and 7,979 patients
- Modified care team billing pipeline to fix two bugs: minutes logged on same day by two members count for only one (69,502 records), only one program (RPM/CCM) in reports for each day (546 records); validated with sample-based statistical tests
- Redesigned daily program enrollment report into per-clinic breakdown table; migrated script into production codebase

International Fragrances & Flavors June 2025 – Jan 2026
Engineering Co-Op – Computer Science (Protein Products Manufacturing Division) Gibson City, IL

- Architected a Python ETL pipeline extracting signals from Aspen SQL MES databases and parsing raw XML into DataFrames, running ~2x faster than Excel-based workflows; built as a Seeq-independent fallback for Python analysis
- Built a real-time Seeq dashboard using a Self-Organizing Map and ARIMA-based time-series analysis to flag low water-pressure events from flow, pressure, and feed-rate signals; worked with operations to diagnose root causes (pipe leak, modified piping)
- Applied signal processing and statistical modeling (autoregression, linear regression) in Python to diagnose process failures, e.g., traced flour-bin infeed shutdowns to brief negative-weight sensor readings hidden by smoothing
- Modeled recycle screw conveyor input-output across process conditions to set optimal motor settings, raising output from 400–800 to 900–1,100 lb/hr (1,000 target) in plant testing and saving an estimated ~\$1,000/day in feed
- Debugged PLC logic across Siemens 505 (dryer) and RSLogix (extruder) systems: fixed inter-PLC slidegate communication and status-bit logic, grinder startup sequencing, and aspiration alarms to remove 80% of unplanned dryer/grinder error

PROJECTS & OPEN SOURCE

Cooking Chatbot | *PyTorch, CLIP, PostgreSQL, Google OAuth* **cookingchatbot.vercel.app**

- Fine-tuned OpenAI's CLIP vision transformer (PyTorch, WebDataset sharding; 597 classes × 250 images) with a prompt ensemble pipeline, raising zero-shot top-5 accuracy from 45% to 90% on held-out data (~100 classes, 50 images each)
- Developed data cleaning pipelines to normalize 200,000 raw ingredients into 597 canonical classes (substring grouping, manual review of every class) and filter a 2 million-row Kaggle recipe dataset to 1.5M+ usable recipes
- Implemented PostgreSQL (Aiven) backend on Render for 1.5M+ recipes, ingredient lookups, and user preferences that drive recommendations (e.g., no meat for vegetarians), with Google OAuth; models hosted on Hugging Face

Fake News Detection System | *FastAPI, XGBoost, scikit-learn, Hugging Face* **fake-news-detector-sc.vercel.app**

- Deployed serverless misinformation detector (FastAPI/Vercel): SGD-SVM with TF-IDF n-grams on 49K articles (ISOT/DataFlair, ~97% acc.) and XGBoost with 768-dim MPNet sentence embeddings on 134K tweets (0.89 ROC-AUC)
- Diagnosed data leakage (replies to one statement in both train and test) inflating XGBoost ROC-AUC to 0.99; redesigned evaluation with GroupShuffleSplit CV across 1,058 unique statements, validating a leak-free 0.89 AUC on 130K+ tweets
- Engineered FastAPI routing pipeline: scrapes URLs with 200+ chars of text (BeautifulSoup), routes X/Twitter links to the tweet model (Twitter APIs), builds 46 lexical/NER features and MPNet embeddings; ~2s/article, ~0.5s/tweet when warm

scikit-learn | *Open-Source Contributor (Documentation)* **github.com/scikit-learn/scikit-learn**

- Improved the Jaccard score docstring in the metrics module (#34645)
- Fixed a broken link to the scikit-learn-intelex documentation on the installation page (#34640)

ACTIVITIES

Association for Computing Machinery at UIUC Aug 2026 - Present
Marketing Chair Urbana, IL

- Manage social media strategy for 2,000+ followers, reaching 80,000+ views/month to drive ACM participation and retention

CS 225 - Data Structures Aug 2026 - Present
Course Assistant Urbana, IL

- Assisted with teaching, management, and learning of data structures in lab and office hours settings.